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Exploring Reciprocity Test in Surface Waves: Implications for 1D Velocity Inversion

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Abstract

This study explores the use of reciprocity in surface wave inversion, a concept more commonly applied in traveltimes tomography. We compare 1D shear-wave velocity models from reciprocal shots recorded on the same seismic line to test how consistent the results are. Using field data from 121 geophones and two shots placed on opposite ends of the array, we extract and invert dispersion curves to produce velocity models. The results from both shots show good agreement, especially at greater depths, confirming that reciprocity can help validate inversion results. When differences occur, they are usually caused by noise, near-surface complexity, or overlapping wave modes. These findings suggest that while 1D inversion is reliable under simple conditions, more advanced methods like 2D inversion may be needed in complex areas.

Introduction

Seismic inversion techniques are crucial in characterizing subsurface structures for various geophysical applications, including hydrocarbon exploration, mining, groundwater studies, and engineering site investigations (Virieux & Operto, 2009; Sheriff & Geldart, 1995). Among these techniques, the analysis of surface waves has proven particularly valuable in near-surface geophysical investigations due to their sensitivity to shallow subsurface properties (Park et al., 1999; Socco & Strobbia, 2004).

Surface waves, primarily Rayleigh waves, are characterized by their dispersive nature, meaning that their velocity varies with frequency as a function of subsurface stiffness and stratification. This dispersion can be inverted to derive shear-wave velocity profiles, providing valuable insights into near-surface geology (Xia et al., 1999). Surface wave inversion methods, such as Multichannel Analysis of Surface Waves (MASW) and Spatial Autocorrelation (SPAC), have been widely used in geotechnical site characterization and mineral exploration (Foti S., 2014; Park et al., 2007).

The principle of reciprocity, traditionally employed in seismic traveltimes tomography, states that seismic recordings obtained by interchanging source and receiver locations should yield identical ray paths since ray paths from both directions go through the same subsurface velocity model under ideal conditions (Knopoff & Gangi, 1959; Lay & Wallace, 1995). While extensively validated in seismic tomography using first

arrivals and reflected waves (Aki & Richards, 2002; Sheriff & Geldart, 1995), reciprocity has rarely been tested in surface wave inversion (Knopoff & Gangi, 1962; Xia et al., 2020; Hosseini, 2021).

Given the increased use of surface waves in geotechnical and exploration geophysics, assessing the applicability of reciprocity to surface wave inversion could provide a significant methodological enhancement. Such validation would improve confidence in surface wave inversion results, especially for shallow subsurface characterizations where accuracy is critical.

This study aims to explore the application of the reciprocity principle to surface wave inversion, validating the approach through direct comparison of velocity models derived from reciprocal seismic shots. By doing so, we aim to quantify the reliability and limitations of 1D inversion models obtained using surface wave dispersion data under different geological conditions.

Theory and Methodology

Reciprocity Principle

The reciprocity principle in seismic exploration indicates that if the positions of the source and receiver are interchanged, the resulting seismic recordings should, theoretically, be identical, provided the subsurface properties remain unchanged (Knopoff & Gangi, 1959; Aki & Richards, 2002). This principle has been predominantly applied in traveltimes tomography, using first arrivals and reflected seismic waves. Reciprocity helps verify the consistency of seismic data and can identify errors or inconsistencies related to source and receiver placements or data acquisition procedures.

Data Acquisition and Processing

The field data (Figure 1) were collected using a seismic array consisting of 121 single-component (1C) receivers arranged linearly at intervals of 1 m, while shot points were spaced at intervals of 2 m. An accelerated 200-lb weight drop was employed as the seismic source. To enhance the signal-to-noise ratio (SNR), each shot location was stacked twice, and these stacked signals were subsequently summed during preprocessing to improve the clarity and quality of first arrivals.

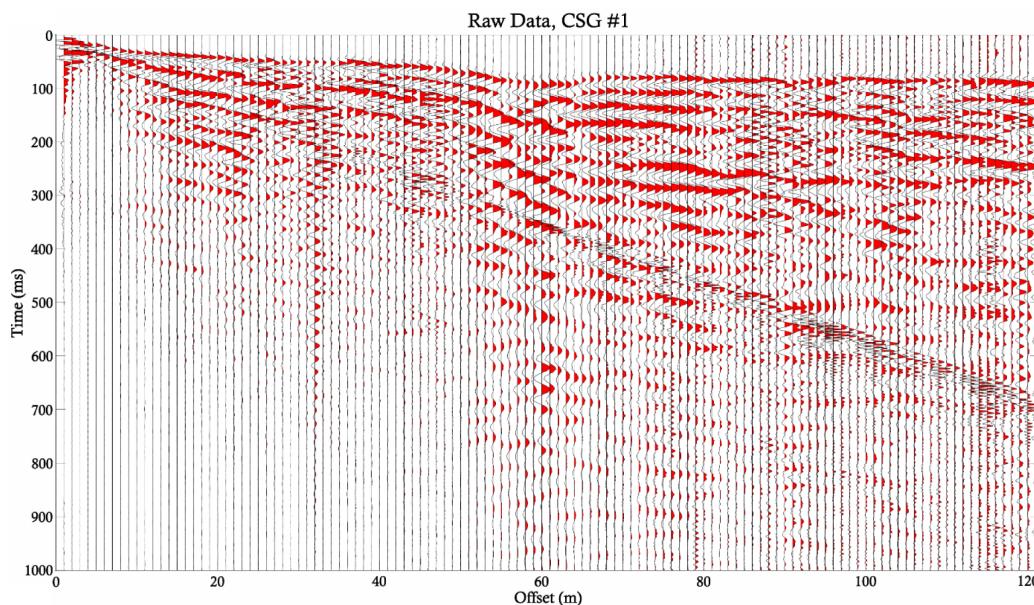


Figure 1—Example of the recorded seismic data; CSG no. 1 showing a very high SNR; hence, no filter is applied.

Once the data was acquired, specific subdatasets and 1D inversion using the Multichannel Analysis of Surface Waves (MASW) is applied. Shot gathers were selectively extracted, consisting of 75 traces to the

right with trace interval of 1 m (Figure 2a) and 75 traces to the left with trace interval of 1 m (Figure 2b) of each shot point S1 and S2 including 5 m offset from the shot. Importantly, the same 75 receiver traces were utilized for both left and right shot gathers to ensure consistency and facilitate direct comparison of reciprocal seismic data. Dispersion curves for the fundamental mode of Rayleigh waves were then accurately extracted from these preprocessed Common Shot Gathers (CSG) through frequency-wavenumber (f-k), and frequency-velocity (f-v) analysis.

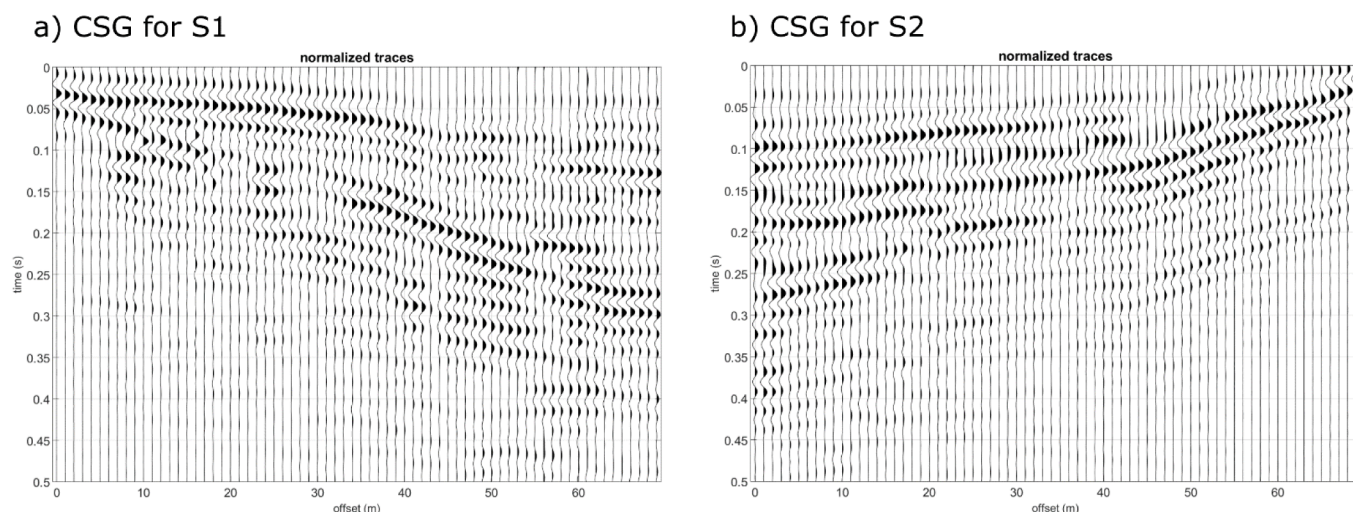


Figure 2—Example of extracted subdataset CSG after applying a low-bandpass filter (0–30 Hz). a) Extracted CSG for reciprocal shot S1. b) Extracted CSG for reciprocal shot S2.

Surface Wave Inversion Method

The Surface Wave Inversion (SWI) analysis was initially performed on synthetic seismic data generated using a Matlab code employing a 2D elastic wave propagation simulation based on a staggered-grid finite-difference method. The synthetic data were generated using a source wavelet with a peak frequency of 10 Hz. Commercial MASW software was employed for subsequent processing steps.

During MASW processing for both synthetic and real data, shot gathers were analyzed by isolating and extracting only the ground-roll component. Phase velocity dispersion curves were calculated over a frequency range of 5 Hz to 30 Hz, considering a shear-wave velocity range between of 500 m/s and 3000 m/s for the synthetic data, and a shear-wave velocity range between of 100 m/s and 3000 m/s for the real data. Following dispersion curve generation, automatic picking procedure were applied to accurately identify fundamental and higher modes.

The final inversion process to derive 1D shear-wave velocity profiles was performed to a credible depth of 20 m, which aligns with typical SWI analysis standards. This depth was determined based on the commonly accepted criterion that the depth of reliable velocity estimation is generally around one-third of the profile length, which, in this study, was 75 m. Additionally, an initial offset of 5 m was included in the shot gathers to optimize the inversion accuracy.

Results

Synthetic Data

The synthetic data results were analyzed using a 2D velocity model characterized by a background with a downward velocity gradient and a central trapezoidal anomaly (Figures 3b and 4b). Two distinct scenarios were evaluated: the first corresponds to a region located away from the anomaly (Figure 3c), while the second partially intersects the trapezoidal feature (Figure 4c). For both cases, the analysis includes reciprocal

CSG (Figures 3d–3e and 4d–4e), dispersion curves with automatic mode picking (Figures 3f–3g and 4f–4g), and the resulting 1D shear-wave velocity models (Figures 3h–3i and 4h–4i). The corresponding numerical inversion results, including shear-wave velocity, layer thickness, and depth, are summarized in Tables 1 and 2 for both scenarios.

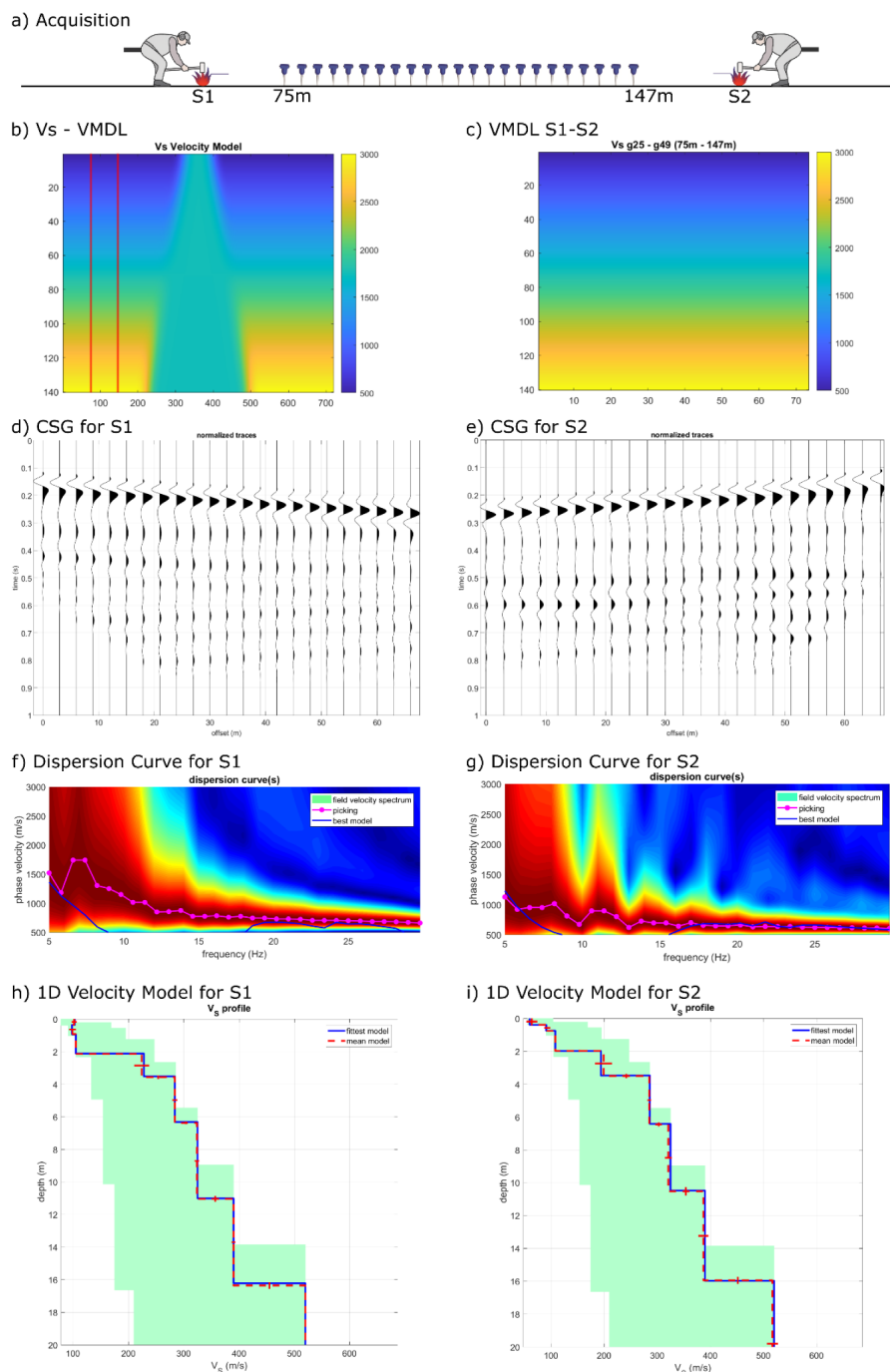


Figure 3—CASE 1 Synthetic Data. Detailed overview of the surface wave inversion (SWI) process using reciprocal shot analysis for synthetic data. (a) Acquisition geometry illustrating two reciprocal shots (S1 and S2) positioned at opposite ends of a seismic profile with 75 m of single-component receivers spaced at 3 m intervals and 6m offset. (b) 2D synthetic velocity model (Vs - VMDL) depicting a velocity gradient with a central trapezoidal anomaly; red vertical lines indicate the locations of the extracted sections analyzed in detail. (c) Extracted velocity model section (VMDL S1-S2) highlighting the analyzed portion of the full model. (d, e) Ground-roll extracted from CSG corresponding to reciprocal shots S1 and S2, respectively. (f, g) Dispersion curves derived from the shot gathers for S1 and S2, showing

automatic mode picking and best-fit models. (h, i) Final 1D shear-wave velocity profiles resulting from the surface wave inversion process for shots S1 and S2, including the fittest model, mean model, and confidence interval (green shading).

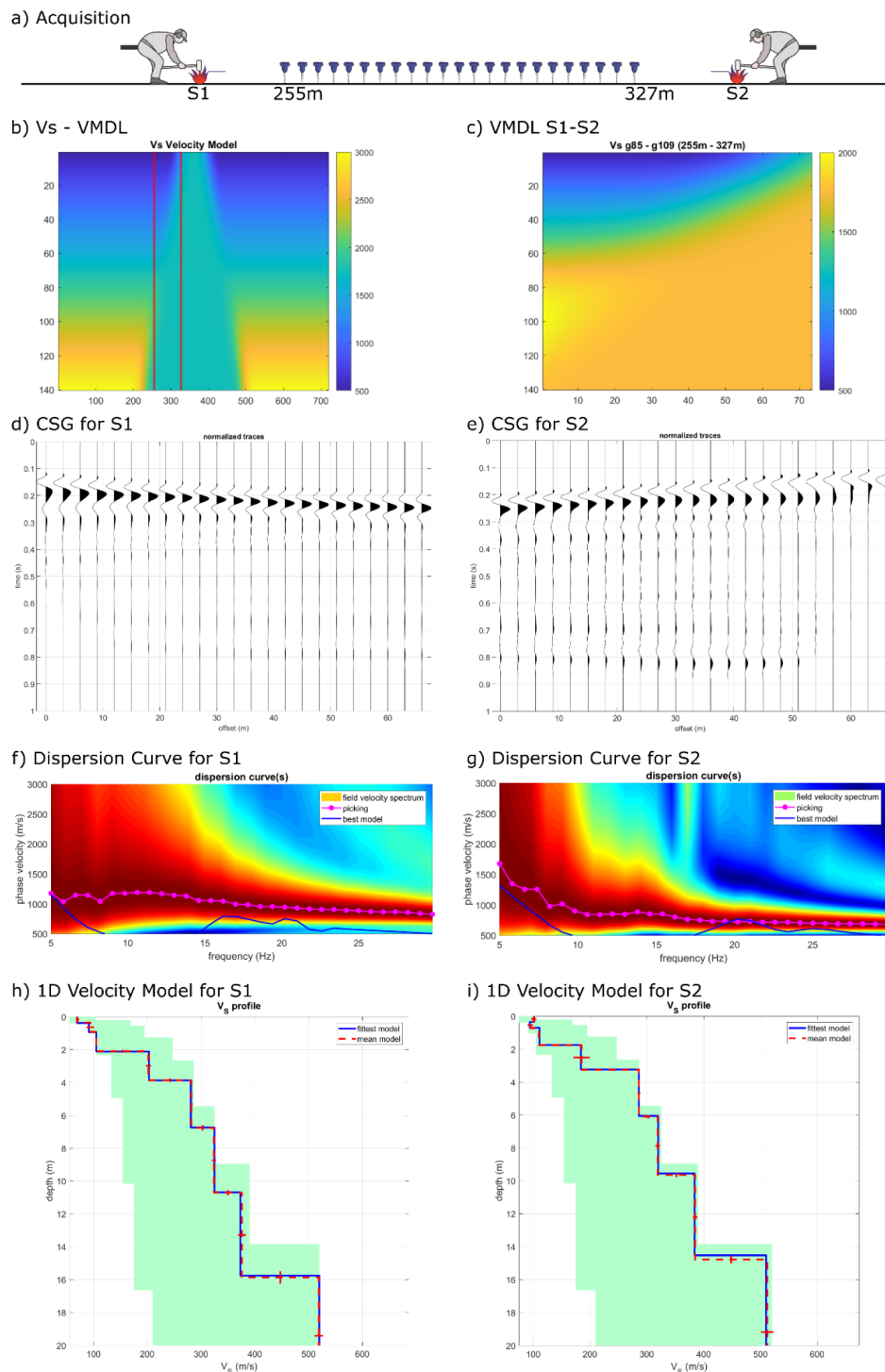


Figure 4—CASE 2 Synthetic Data. Detailed overview of the surface wave inversion (SWI) process using reciprocal shot analysis for synthetic data. (a) Acquisition geometry illustrating two reciprocal shots (S1 and S2) positioned at opposite ends of a seismic profile with 75 m of single-component receivers spaced at 3 m intervals and 6m offset. (b) 2D synthetic velocity model (Vs - VMDL) depicting a velocity gradient with a central trapezoidal anomaly; red vertical lines indicate the locations of the extracted sections analyzed in detail. (c) Extracted velocity model section (VMDL S1-S2) highlighting the analyzed portion of the full model. (d, e) Ground-roll extracted from CSG corresponding to reciprocal shots S1 and S2, respectively. (f, g) Dispersion curves derived from the shot gathers for S1 and S2, showing

automatic mode picking and best-fit models. (h, i) Final 1D shear-wave velocity profiles resulting from the surface wave inversion process for shots S1 and S2, including the fittest model, mean model, and confidence interval (green shading).

Table 1—Layer-by-layer results from 1D shear-wave velocity (Vs) inversion for Case 1 – Synthetic Data. The table presents the velocity, thickness, and cumulative depth (until 20 m) of each layer for two reciprocal shots: Shot 1 (S1) and Shot 2 (S2). These results are derived from surface wave inversion (SWI) using dispersion curves extracted from CSG recorded at the same geophone array.

CASE 1 Synthetic Data							
Shot 1 – S1				Shot 2 – S2			
Layer	Vs (m/s)	Thickness (m)	Depth (m)	Layer	Vs (m/s)	Thickness (m)	Depth (m)
1	103	0.3614	0.3614	1	60	0.3900	0.3900
2	98	0.5772	0.9386	2	91	0.3570	0.7470
3	105	1.1866	2.1252	3	108	1.2389	1.9859
4	228	1.4000	3.5252	4	194	1.4926	3.4785
5	284	2.8000	6.3252	5	286	2.9468	6.4253
6	325	4.6898	11.0150	6	325	4.0614	10.4867
7	390	5.2140	16.2290	7	390	5.4997	15.9864
8	520	7.2110	20.0000	8	520	7.7943	20.0000

Table 2—Layer-by-layer results from 1D shear-wave velocity (Vs) inversion for Case 2 – Synthetic Data. The table presents the velocity, thickness, and cumulative depth (until 20 m) of each layer for two reciprocal shots: Shot 1 (S1) and Shot 2 (S2). These results are derived from surface wave inversion (SWI) using dispersion curves extracted from CSG recorded at the same geophone array.

CASE 2 Synthetic Data							
Shot 1 – S1				Shot 2 – S2			
Layer	Vs (m/s)	Thickness (m)	Depth (m)	Layer	Vs (m/s)	Thickness (m)	Depth (m)
1	69	0.3838	0.3838	1	102	0.3457	0.3457
2	91	0.5553	0.9391	2	94	0.3600	0.7057
3	105	1.1878	2.1269	3	111	1.0420	1.7477
4	203	1.7484	3.8753	4	184	1.4958	3.2435
5	281	2.8702	6.7455	5	286	2.8145	6.0580
6	325	3.9450	10.6906	6	320	3.5000	9.5580
7	373	5.0699	15.7604	7	384	4.9698	14.5278
8	520	7.0000	20.0000	8	510	8.5739	20.0000

Real Data

The real data results were interpreted using a 2D shear-wave velocity model that exhibits a central low to medium velocity anomaly. Three representative scenarios were investigated: two located near the lateral edges of the model (Figures 5b and 7b), and one situated closer to the central anomalous zone (Figure 6b). For each scenario, reciprocal shot gathers were analyzed (Figures 5c–5d, 6c–6d, and 7c–7d), followed by dispersion curve extraction and mode picking (Figures 5e–5f, 6e–6f, and 7e–7f). The resulting 1D shear-wave velocity profiles for each reciprocal pair are presented and visually compared (Figures 5g–5h, 6g–6h, and 7g–7h). The detailed inversion results—including Vs values, layer thicknesses, and cumulative depths—are summarized in Tables 3, 4, and 5 for each respective case.

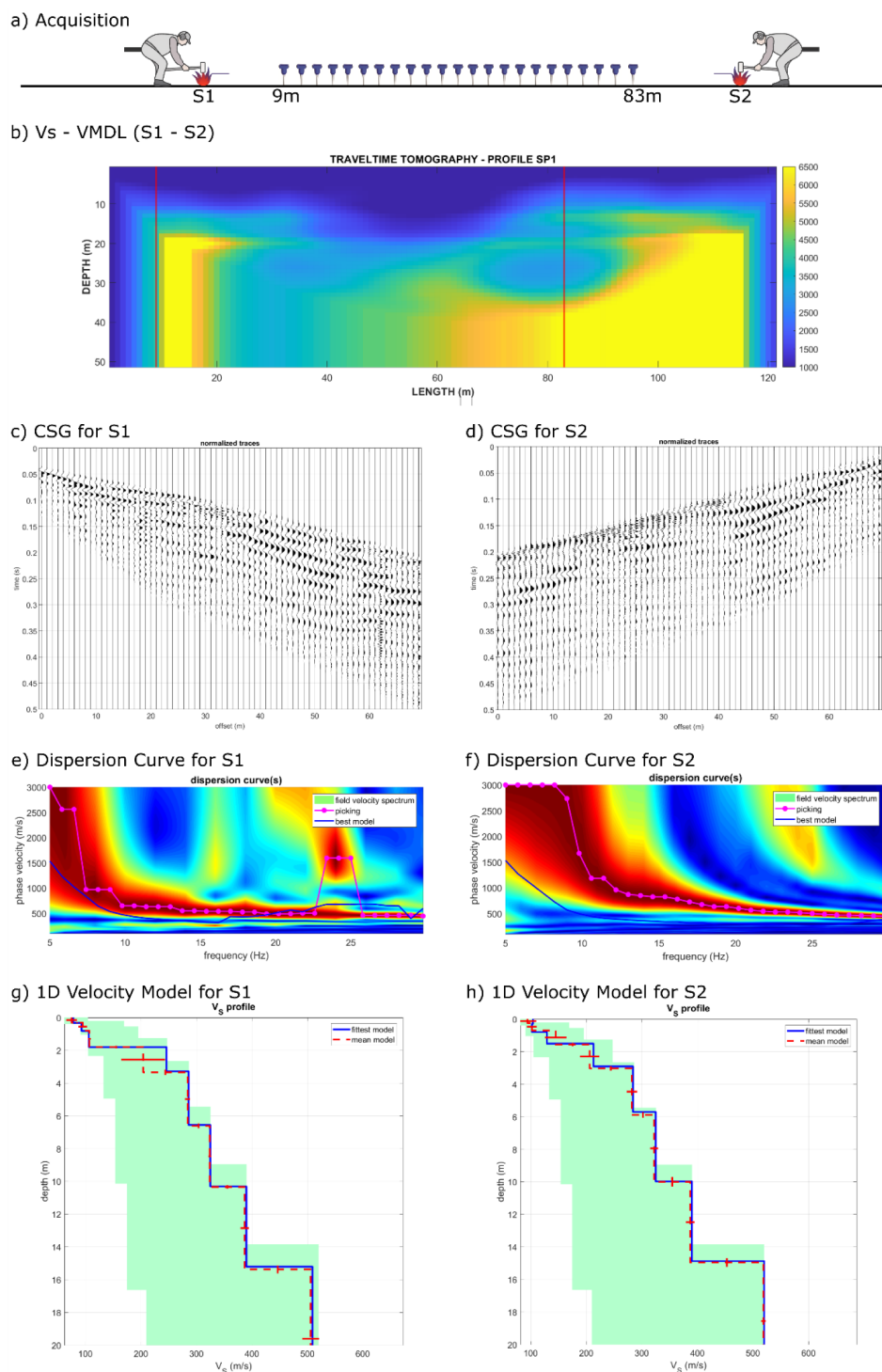


Figure 5—CASE 1 Real Data. Comprehensive presentation of surface wave inversion (SWI) results from reciprocal shot analysis for real seismic data. (a) Acquisition geometry illustrating two reciprocal shots (S1 and S2) positioned at opposite ends of a linear seismic array of 75 m single-component receivers spaced at 1 m intervals with 5 m offset. (b) 2D shear-wave velocity model (V_s - VMDL) obtained from traveltime tomography (TTT) showing a central anomaly of low to medium velocity; red vertical lines denote the location of the extracted sections used for detailed SWI analysis. (c, d) Ground-roll extracted from CSG corresponding to reciprocal shots S1 and S2, highlighting the seismic responses recorded at respective receiver positions. (e, f) Dispersion curves derived from these CSGs, with clearly indicated automatic picking of fundamental modes and corresponding best-fit velocity models. (g, h) Resultant 1D shear-wave velocity profiles for shots S1 and S2, showing the fittest velocity model, mean model, and associated confidence intervals (green shaded area).

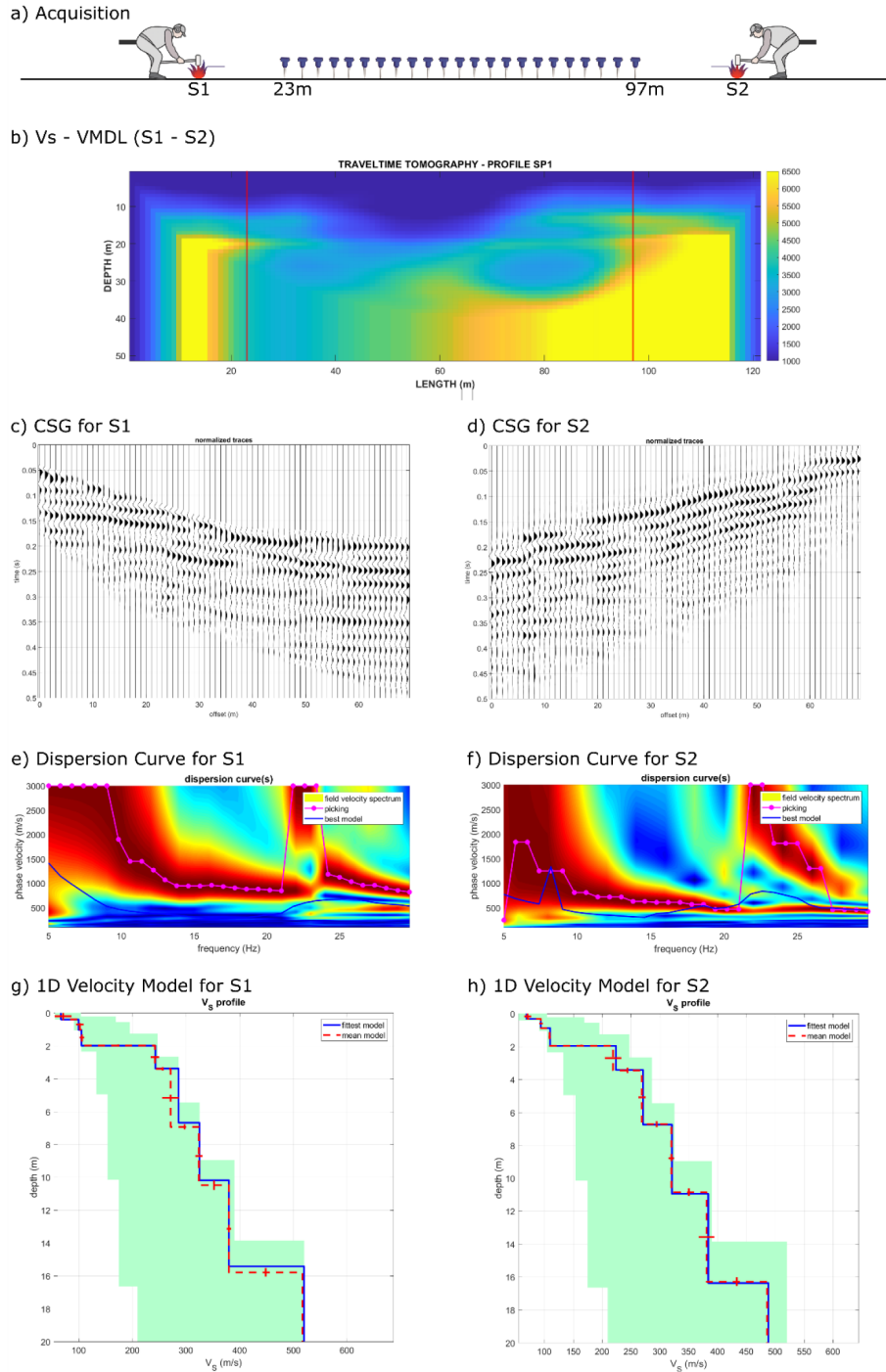


Figure 6—CASE 2 Real Data. Comprehensive presentation of surface wave inversion (SWI) results from reciprocal shot analysis for real seismic data. (a) Acquisition geometry illustrating two reciprocal shots (S1 and S2) positioned at opposite ends of a linear seismic array of 75 m single-component receivers spaced at 1 m intervals with 5 m offset. (b) 2D shear-wave velocity model (Vs - VMDL) obtained from traveltime tomography (TTT) showing a central anomaly of low to medium velocity; red vertical lines denote the location of the extracted sections used for detailed SWI analysis. (c, d) Ground-roll extracted from CSG corresponding to reciprocal shots S1 and S2, highlighting the seismic responses recorded at respective receiver positions. (e, f) Dispersion curves derived from these CSGs, with clearly indicated automatic picking of fundamental modes and corresponding best-fit velocity models. (g, h) Resultant 1D shear-wave velocity profiles for shots S1 and S2, showing the fittest velocity model, mean model, and associated confidence intervals (green shaded area).

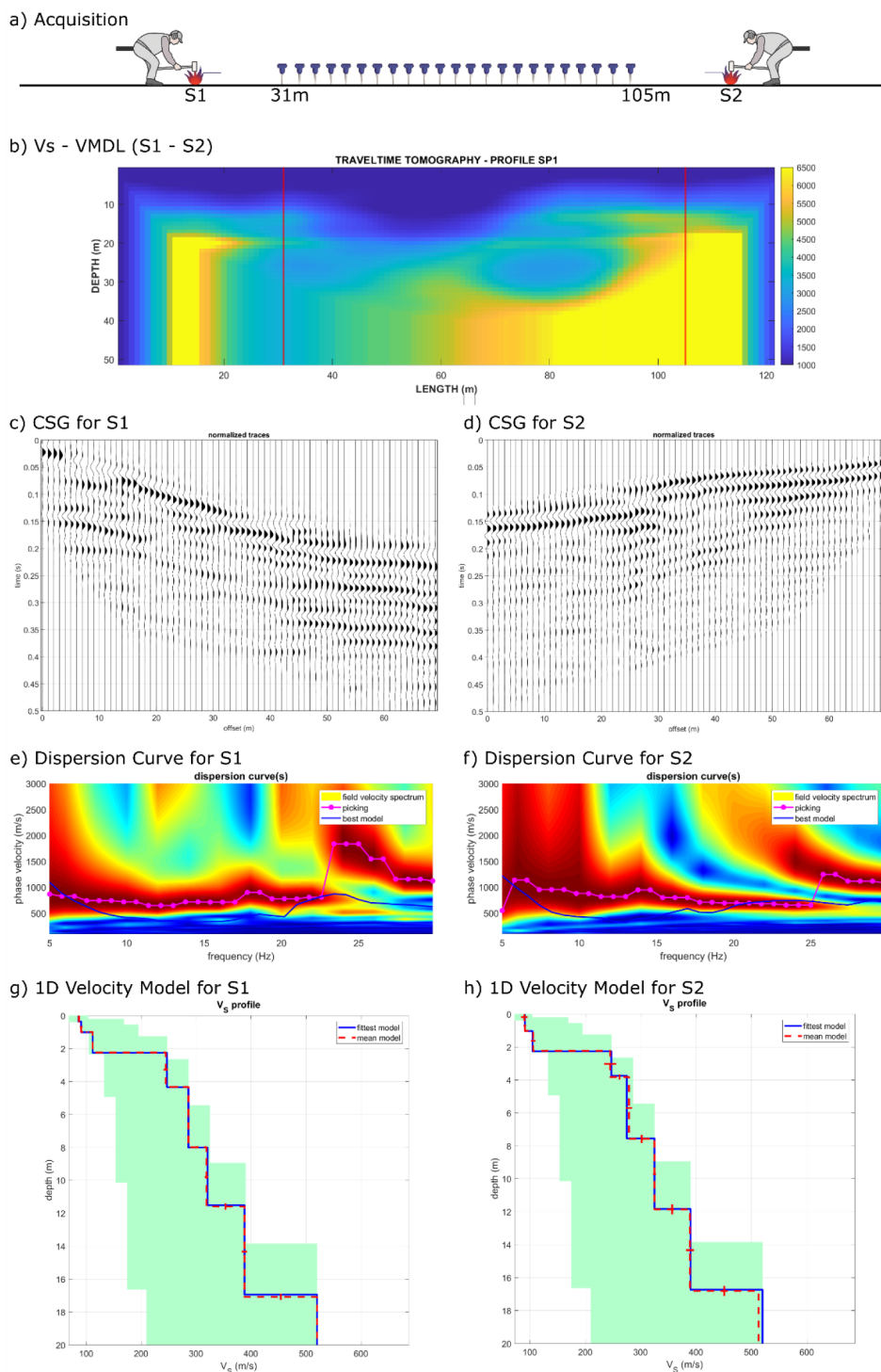


Figure 7—CASE 3 Real Data. Comprehensive presentation of surface wave inversion (SWI) results from reciprocal shot analysis for real seismic data. (a) Acquisition geometry illustrating two reciprocal shots (S1 and S2) positioned at opposite ends of a linear seismic array of 75 m single-component receivers spaced at 1 m intervals with 5 m offset. (b) 2D shear-wave velocity model (Vs - VMDL) obtained from traveltime tomography (TTT) showing a central anomaly of low to medium velocity; red vertical lines denote the location of the extracted sections used for detailed SWI analysis. (c, d) Ground-roll extracted from CCSG corresponding to reciprocal shots S1 and S2, highlighting the seismic responses recorded at respective receiver positions. (e, f) Dispersion curves derived from these CSGs, with clearly indicated automatic picking of fundamental modes and corresponding best-fit velocity models. (g, h) Resultant 1D shear-wave velocity profiles for shots S1 and S2, showing the fittest velocity model, mean model, and associated confidence intervals (green shaded area).

Table 3—Layer-by-layer results from 1D shear-wave velocity (Vs) inversion for Case 1 – Real Data. The table presents the velocity, thickness, and cumulative depth (until 20 m) of each layer for two reciprocal shots: Shot 1 (S1) and Shot 2 (S2). These results are derived from surface wave inversion (SWI) using dispersion curves extracted from CSG recorded at the same geophone array.

CASE 1 Real Data							
Shot 1 – S1				Shot 2 – S2			
Layer	Vs (m/s)	Thickness (m)	Depth (m)	Layer	Vs (m/s)	Thickness (m)	Depth (m)
1	78	0.3292	0.3292	1	104	0.2891	0.2891
2	93	0.4874	0.8165	2	102	0.5087	0.7978
3	106	0.9888	1.8053	3	129	0.7142	1.5120
4	246	1.4810	3.2863	4	213	1.4000	2.9120
5	286	3.2712	6.5575	5	284	2.8000	5.7120
6	325	3.7583	10.3158	6	325	4.2684	9.9804
7	390	4.9000	15.2158	7	390	4.9000	14.8804
8	509	8.3380	20.0000	8	520	7.0000	20.0000

Table 4—Layer-by-layer results from 1D shear-wave velocity (Vs) inversion for Case 2 – Real Data. The table presents the velocity, thickness, and cumulative depth (until 20 m) of each layer for two reciprocal shots: Shot 1 (S1) and Shot 2 (S2). These results are derived from surface wave inversion (SWI) using dispersion curves extracted from CSG recorded at the same geophone array.

CASE 2 Real Data							
Shot 1 – S1				Shot 2 – S2			
Layer	Vs (m/s)	Thickness (m)	Depth (m)	Layer	Vs (m/s)	Thickness (m)	Depth (m)
1	67	0.3900	0.3900	1	69	0.3216	0.3216
2	100	0.6408	1.0308	2	94	0.5513	0.8730
3	105	0.9522	1.9830	3	110	1.0856	1.9586
4	243	1.4000	3.3830	4	224	1.4632	3.4218
5	286	3.2983	6.6813	5	271	3.3170	6.7387
6	325	3.5000	10.1813	6	321	4.2072	10.9459
7	380	5.2414	15.4227	7	384	5.4244	16.3703
8	520	7.7488	20.0000	8	488	8.2965	20.0000

Table 5—Layer-by-layer results from 1D shear-wave velocity (Vs) inversion for Case 3 – Real Data. The table presents the velocity, thickness, and cumulative depth (until 20 m) of each layer for two reciprocal shots: Shot 1 (S1) and Shot 2 (S2). These results are derived from surface wave inversion (SWI) using dispersion curves extracted from CSG recorded at the same geophone array.

CASE 3 Real Data							
Shot 1 – S1				Shot 2 – S2			
Layer	Vs (m/s)	Thickness (m)	Depth (m)	Layer	Vs (m/s)	Thickness (m)	Depth (m)
1	86	0.3662	0.3662	1	91	0.3900	0.3900
2	91	0.6489	1.0151	2	91	0.6462	1.0362
3	112	1.2448	2.2599	3	105	1.2344	2.2706
4	247	2.0933	4.3532	4	247	1.4837	3.7543
5	286	3.6672	8.0204	5	275	3.8042	7.5585
6	321	3.5000	11.5204	6	325	4.2839	11.8424
7	388	5.4391	16.9595	7	390	4.9000	16.7424
8	520	8.5330	20.0000	8	520	9.8884	20.0000

Discussion

Interpretation of Results

Synthetic Data. The application of the reciprocity principle to surface wave inversion produced encouraging results, particularly in the synthetic scenarios. In Case 1 (Figure 3), using a gradient background velocity model with a central trapezoidal anomaly, the reciprocal dispersion curves and 1D velocity profiles were found to be very similar. Differences between the two models were most pronounced in the upper few meters of the profile (Figures 3h and 3i), as observed in the velocity Table 1. These discrepancies may be due to the limitations imposed by the upper frequency bound of 30 Hz, potential boundary effects near the model edges, or subtle influences from the anomaly itself. Nevertheless, from layers 5 to 8, both inversions yielded nearly identical Vs values, confirming strong reciprocity at depths around 20 m.

In Case 2 (Figure 4), where the analysis was conducted closer to the trapezoidal anomaly, the reciprocal 1D velocity models still showed very good agreement. While there was slightly more variation in the velocity values compared to Case 1—especially in the near-surface layers—the overall match between the two reciprocal inversions remained satisfactory both qualitatively (Figure 4h and 4i) and quantitatively (Table 2). The current results reinforce that even in the presence of subsurface anomalies, the reciprocity principle holds well within the depth range of interest.

Additionally, automatic mode picking was used in all cases to eliminate interpreter bias and ensure consistent conditions across reciprocal shot pairs. This approach reinforced the validity of comparing inversion results. It is also important to note that a separate synthetic case involving a purely gradient model (not shown in the main figures) yielded nearly identical dispersion curves and velocity profiles between reciprocal shots, suggesting that under ideal, noise-free, and homogeneous conditions, reciprocity in surface wave inversion holds exceptionally well.

Real Data. The real data results demonstrated patterns generally consistent with those observed in the synthetic cases (Figures 5, 6, and 7). Ground roll identification and separation were relatively straightforward. Across all three real-data cases, dispersion curves consistently showed the presence of higher modes that may have overlapped with the fundamental mode (Figures 5e-5f, 6e-6f, and 7e-7f), potentially introducing minor discrepancies. Nevertheless, because these higher modes were similarly present in both reciprocal shots, the automatic picking process was retained for consistency.

In Case 1 (Figure 5, Table 3), minor misfits were observed primarily in the upper ~4 meters (Figures 5g and 5h), where a higher mode appears to affect the dispersion curves around 25 Hz (Figure 5e). In Case 2 (Figure 6, Table 4), the presence of a more pronounced higher mode in the 20–30 Hz range was noted (Figures 6e and 6f); however, both shots exhibited similar higher-mode characteristics, resulting in closely matching 1D velocity models at shallow depths (Figures 6g and 6h). Below 4 meters, slight differences emerge due to a variation in the fundamental mode between 5–10 Hz (Figures 6e and 6f). Case 3 (Figure 7, Table 5) yielded the most consistent results between reciprocal shots throughout the entire depth range of the 1D profiles (Figures 7g and 7h), supporting the method's robustness.

RMS Error. The RMS error analysis provides a quantitative measure of similarity between reciprocal 1D velocity models and supports the overall qualitative assessment. In Synthetic Case 1 (Figure 8a), the RMS difference between interpolated velocity profiles from reciprocal shots is only 11.86 m/s, corresponding to a 3.84% relative error. This low RMS highlights the strong agreement between reciprocal inversions in a region away from the anomaly.

In Synthetic Case 2 (Figure 8b), where the analysis intersects part of the central anomaly, the RMS error increases to 18.03 m/s, or 6.00%. This increase is expected due to the local heterogeneity introduced by the anomaly, which affects wave propagation paths and causes slight discrepancies in inversion outcomes.

Despite this, the results still exhibit a high degree of similarity, confirming that reciprocity holds well even in regions of moderate complexity.

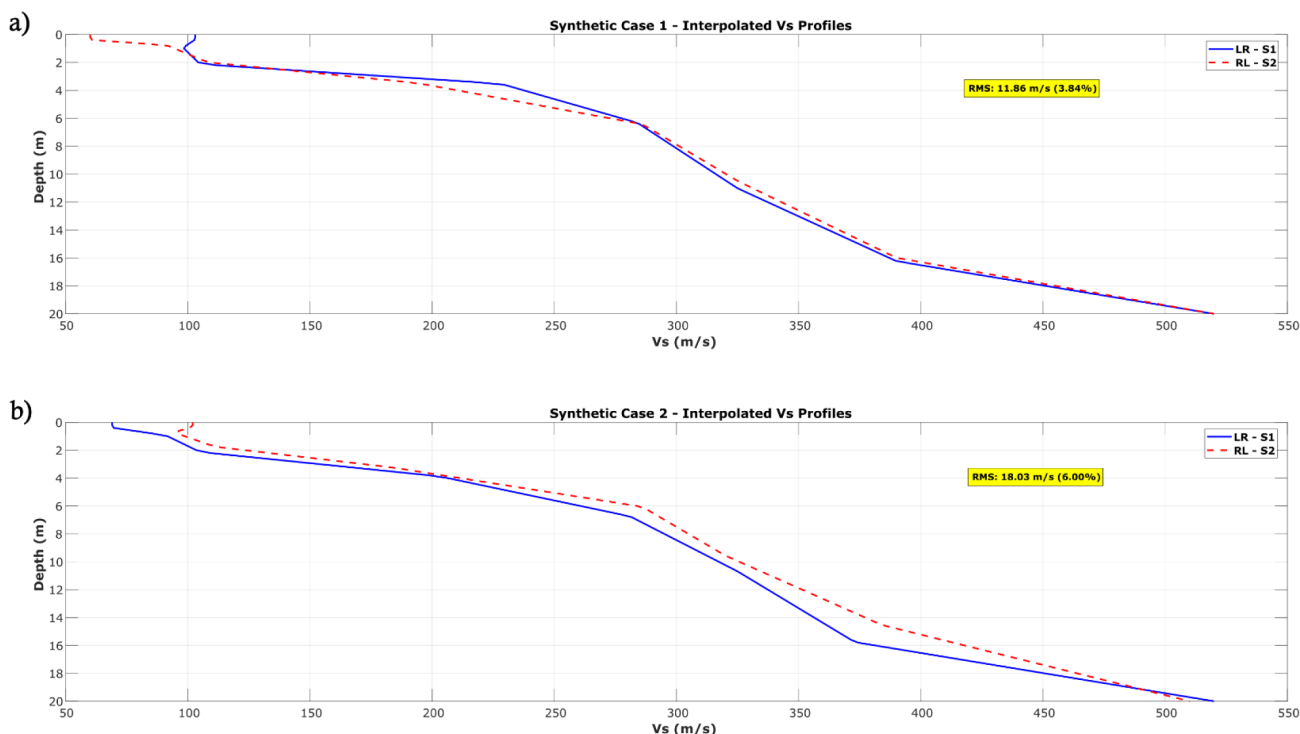


Figure 8—Interpolated 1D shear-wave velocity (V_s) profiles for reciprocal shots in (a) Synthetic Case 1 and (b) Synthetic Case 2. The solid blue line represents the Left-to-Right (LR) shot, while the dashed red line corresponds to the Right-to-Left (RL) shot. The profiles are plotted to a depth of 20 meters. RMS error values between the profiles are shown in yellow boxes, both in absolute terms and as a percentage of the average velocity over depth. Synthetic Case 1 shows a low RMS of 11.86 m/s (3.84%), indicating a very high degree of consistency. Synthetic Case 2, conducted closer to the trapezoidal anomaly, results in a slightly higher RMS of 18.03 m/s (6.00%), but still within acceptable limits for validating reciprocity.

These RMS values validate the robustness of the inversion methodology and the effectiveness of using automatic picking to maintain consistency. Furthermore, they provide a baseline for comparison with real data RMS values, which are expected to be slightly higher due to additional field-related uncertainties.

RMS error values generally corroborated the qualitative observations. Synthetic models produced lower RMS errors overall, particularly in ideal gradient cases. In real data, slightly elevated RMS values were recorded, corresponding to zones influenced by overlapping higher modes or increased noise. The close alignment of RMS trends with the visual and numerical consistency of 1D models reinforces the reliability of the surface wave inversion under the reciprocity framework.

The RMS error analysis for the real data further substantiates the qualitative assessment of reciprocity. In Real Case 1 (Figure 9a), which is located near the edge of the profile, slight dispersion curve distortions from higher modes around 25 Hz were noted. Nevertheless, the RMS error remains low at 11.78 m/s (3.72%), confirming that overall agreement between velocity profiles remains strong. In Case 2 (Figure 9b), located closer to the center of the profile, higher mode interference is more prominent between 20–30 Hz. The RMS rises modestly to 16.99 m/s (5.42%), indicating slightly increased deviation due to variations in the fundamental mode and greater subsurface complexity.

Real Case 3 (Figure 9c), also positioned toward the profile edge, stands out with the lowest RMS error among the real data cases—7.84 m/s (2.64%)—demonstrating exceptional consistency in reciprocal velocity profiles. This supports the earlier observation that Case 3 exhibited the most similar velocity models along the full depth range.

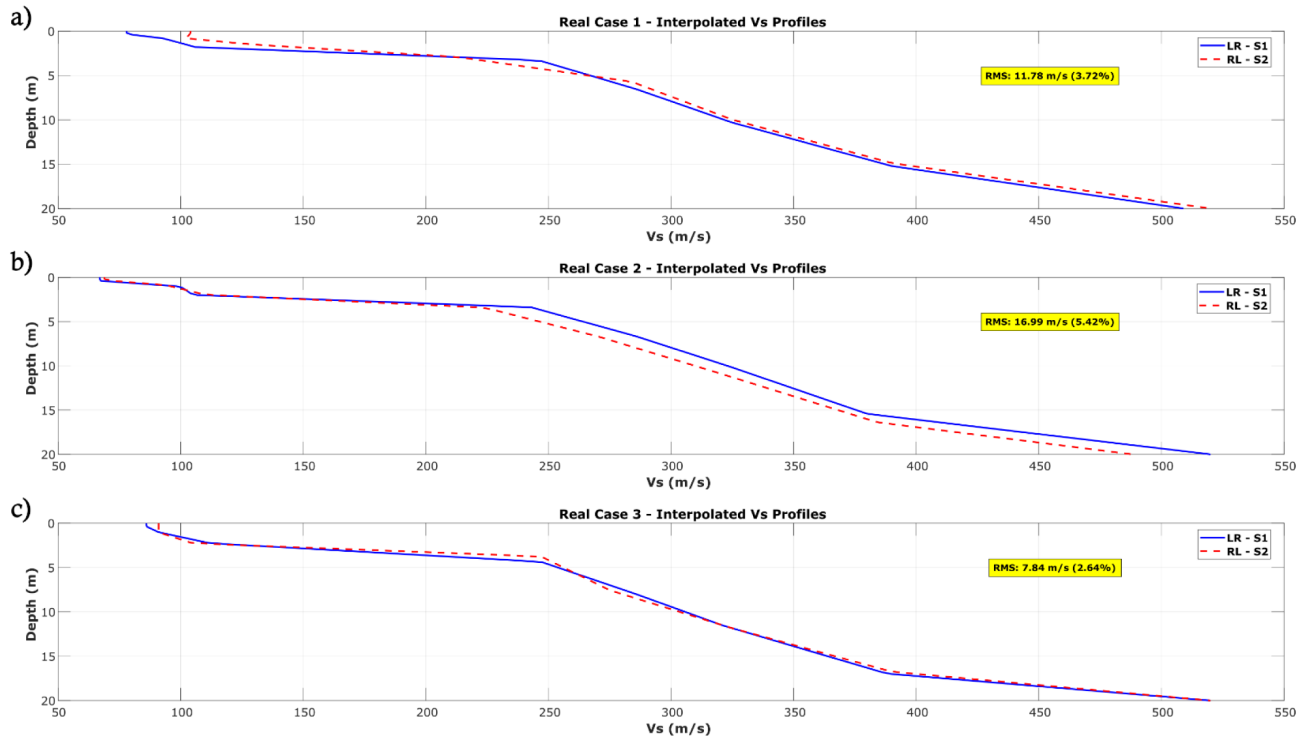


Figure 9—Interpolated 1D shear-wave velocity (V_s) profiles for reciprocal shots in (a) Real Case 1, (b) Real Case 2, and (c) Real Case 3. The blue line represents the Left-to-Right (LR) shot and the red dashed line the Right-to-Left (RL) shot. The RMS error values are displayed on each subplot as absolute values and percentages. Real Case 1 shows an RMS of 11.78 m/s (3.72%), Case 2 has 16.99 m/s (5.42%), and Case 3 yields the lowest RMS of 7.84 m/s (2.64%). These metrics reflect good alignment between reciprocal velocity profiles across all cases.

Together with the synthetic data RMS results, these real-case values confirm that the inversion framework is both robust and reliable under realistic conditions. They also reinforce the conclusion that even when higher modes are present or data quality varies, the automatic picking and processing methodology enables consistent recovery of subsurface structure from reciprocal shot pairs.

Implications for Future Studies

These findings highlight the potential and limitations of 1D surface wave inversion in reciprocal shot analysis. In relatively homogeneous or gently varying subsurface conditions, 1D inversion appears suitable and robust, with the reciprocity principle providing a useful internal validation tool.

However, in cases of complex geology or lateral heterogeneities, the 1D assumption may oversimplify subsurface behavior, leading to mismatches in reciprocal velocity models. In such settings, transitioning to more advanced 2D surface wave inversion or full-waveform techniques is advisable. These methods offer better resolution of structural complexity and can more accurately characterize sites with strong lateral variations.

Future studies should consider incorporating reciprocity-based validation as a standard quality check in near-surface geophysical surveys. Additionally, combining vertical and horizontal components of Rayleigh waves and improving mode discrimination techniques could further enhance the reliability of surface wave inversion results.

Conclusions

This study demonstrates the feasibility and reliability of applying the reciprocity principle to surface wave inversion. Through reciprocal shot analysis on both synthetic and real seismic datasets, 1D shear-wave

velocity models showed high degrees of consistency when similar acquisition geometries and processing methodologies were used.

In the synthetic models, particularly under ideal conditions and in regions of moderate complexity, reciprocal inversions exhibited minimal discrepancies, with RMS errors remaining under 6%. Even in the presence of a subsurface anomaly, inversion results remained comparable, affirming the robustness of the approach.

Real data scenarios, although inherently more variable, further validated the methodology. Across three test cases, RMS differences remained low (below 5.5%), and most variations were attributable to the influence of higher modes, noise, or shallow-layer complexities. Cases located near the profile edges showed particularly strong agreement, while the central case exhibited slightly greater variability.

The integration of automatic mode picking and a consistent acquisition strategy helped reduce interpreter bias and ensure reliable comparative outcomes. The RMS metrics, both qualitative and quantitative, served as a valuable diagnostic tool for assessing inversion quality and identifying deviations.

Ultimately, the study confirms that 1D surface wave inversion using reciprocal shot geometry is a valid and effective technique for near-surface characterization—especially in environments where full 2D or 3D inversion may not be feasible. Incorporating reciprocity checks can enhance data validation procedures, and future applications may benefit from expanding this approach to multi-component and multi-mode analyses.

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